

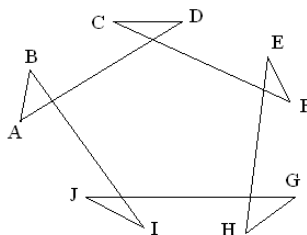
JMO – 2013

Answer all questions. All questions carry equal marks.

Full Marks 100

1.

(a) From the figure $\angle A + \angle B + \angle C + \angle D + \angle E + \angle F + \angle G + \angle H + \angle I + \angle J = ?$



(b) Find the value of n such that $(10^n - 1)(10^n + 1) = 99999999$.

2.

(a) What is the unit digit in 2013^{2015} ?

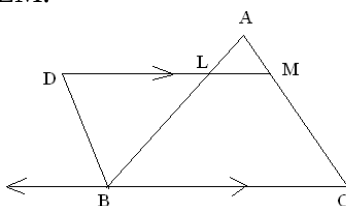
(b) Find the sum $\frac{1}{2 \times 5} + \frac{1}{5 \times 8} + \frac{1}{8 \times 11} + \frac{1}{11 \times 14} + \dots$ upto 20 terms.

3. Prove that $3333^{4444} + 4444^{3333}$ is divisible by 7.

4. Prakash multiplied 253 with a number and got 34426. If both the 4's in the product are wrongly written then what is the correct product?

5. 108 marbles were to be given to A, B and C. At first A got the highest number of marbles and C got the least number of marbles. 'A' then gave some marbles to B and C that doubled the marbles they had plus another 4 marbles each. 'B' then gave some marbles to A and C that doubled the number of marbles they had plus another 4 marbles each. Finally C gave some marbles to A and B that 4 more than doubled the number of marbles they had. Now they had the same number of marbles. How many marbles did each of them get at first?

6. In the figure, CD and BD are the bisectors of the interior angle $\angle C$ and exterior angle $\angle B$ of triangle ABC. They meet each other at D. DM is parallel to BC and meets AB at L. If $LB = 5$ cm and $CM = 7$ cm, Find length of LM.



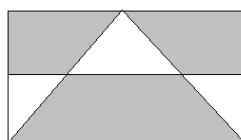
7. The sum of any three consecutive digits of a 10 digits number is 18. Determine the value of N , if the number is denoted by $N * 8 * * * * 6 * 4$.

8. A merchant gets 4% profit even after giving a discount of 20% on list price. By what percent the list price is more than the cost price?

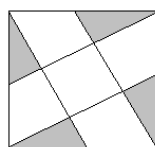
9. The admission ticket to visit the museum is Rs. 6/-, when a concession is given the number of museum goers increased by 50%. The new revenue each day increased by 25%. How much is one concession ticket?

10. Four digit numbers are formed using the digit 1, 2, 3, 4 such that no digits repeated in any number. How many of these numbers are divisible by 11?

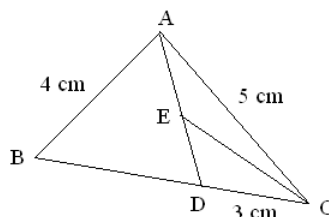
11. Anjana rolls one ordinary die repeatedly, keeping the record of each number she rolls and stopping as soon as any number is rolled for the third time. If Anjana stops after her twelfth roll, and sum of the numbers obtained with these rolls is 47, which number has occurred three times?
12. $\underbrace{111\dots111}_{2002 \text{ digits}} \underbrace{555\dots555}_{2002 \text{ digits}}$ is the product of two consecutive odd numbers. Find the sum of these two odd numbers.
13. Starting from number 1 all consecutive numbers up to 5000 are written on a straight line to form a new number. What is the 1992nd digit of the new number?
14. What fraction of the diagram is shaded?



15. The top left corner of the given square is joined to the mid point of the bottom edge, the mid point of the top edge is joined to the bottom right corner; the top right corner is joined to the midpoints of the left edge, and the mid point of the right edge is joined to the bottom left corner. What fraction of square is shaded?



16. A train, while travelling from station P to station Q, stops for 30 minutes after 1 hour of its start, because of repair of some fault found in the engine. Then it continues moving at fourfifth of its usual speed and arrives 2 hours late at Q. Had the train covered 80 kms, more than it actually did before the fault was found with the engine, it would have been just 1 hour late in arriving Q. Find the usual speed of the train.
17. A cheque is written for 'x' rupees and 'y' paisa where 'x' and 'y' each is non-zero numbers. By mistake it was cashed 'y' rupees and 'x' paisa. If the incorrect amount exceed the correct amount by 17 rupees and 82 paisa, then determine the possible pairs of the values of 'x' and 'y'.
18. If x men working x hours a day for each x days produces x articles, then find out the number of articles (not necessarily integers) produced by y men working y hours a day for each of y days?
19. In $\triangle ABC$, AD bisects $\angle BAC$ and meets BC at D and CE bisects $\angle ACB$ and meets AD at E. If AB = 4 cm, AC = 5 cm, CD = 3 cm and area of the triangle ABC is 18cm^2 . Find the area of the triangle AEC.



20. In the following addition what digit does each letter represent?

$$\begin{array}{r}
 B A A B B \\
 + B A A B B \\
 \hline
 A A C C C
 \end{array}$$

